Demonstration of Proposed Features

Date	15 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters – AI-Based Flood Prediction System
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document summarizes the features proposed during the planning phase of the **Rising Waters – AI-Based Flood Prediction System** and records their implementation and demonstration status. It serves as evidence that the project objectives were successfully achieved and that the proposed solution was effectively developed and demonstrated.

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S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	User-Friendly Prediction Interface	Developed a responsive web interface using Flask, HTML, CSS, and JavaScript to collect environmental parameters from users.	Implemented	Yes	Successfully demonstrated during project review.
2	AI-Based Flood Prediction	Implemented the XGBoost machine learning model to predict flood risk using environmental and historical flood data.	Implemented	Yes	Prediction accuracy validated with the Kaggle dataset.
3	Data Validation Module	Added server-side validation to ensure accurate and complete user inputs before prediction.	Implemented	Yes	Prevents invalid or incomplete submissions.
4	MySQL Database Integration	Integrated MySQL to store prediction records, user inputs, and historical prediction data.	Implemented	Yes	Data storage and retrieval demonstrated successfully.
5	Prediction Results Dashboard	Developed a results page displaying flood risk predictions and related information in a user-friendly format.	Implemented	Yes	Dashboard displayed correctly during demonstration.

6	Prediction History Management	Enabled storage and retrieval of previous prediction records for analysis and future reference.	Implemented	Yes	Successfully demonstrated using sample prediction records.

Feature Implementation Summary:

Metric	**Value**
Total Features Proposed	**6**
Total Features Implemented	**6**
Total Features Demonstrated	**6**

Demonstration Summary:

The **Rising Waters – AI-Based Flood Prediction System** successfully implemented and demonstrated all planned core features within the project timeline. The integration of the Flask web application, MySQL database, and XGBoost machine learning model resulted in a fully functional system capable of predicting flood risk based on environmental parameters. During the demonstration, all features operated as expected, confirming that the project met its functional objectives and was ready for academic evaluation.